Status	Finished
Started	Saturday, 17 October 2026, 10:06 AM
Completed	Saturday, 17 October 2026, 10:17 AM
Duration	11 mins
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution

b. Extraction

c. Selection

d. Insertion

The correct answer is:
Insertion

Question **2**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method

b. By sorting data in a single pass

c. By comparing elements sequentially

d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **3**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It slows down the process
- b. It complicates the selection of items
- c. It makes it easier to select items based on their relationship to the rest of the items
- d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **4**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Rearranging data without sorting
- b. Sorting data sequentially
- c. Dividing the input into parts, solving each part, and combining the solutions
- d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Linear Search
- c. Binary Search
- d. Merge Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Bubble sort
- b. Heap sort
- c. Insertion sort
- d. Quick sort

The correct answer is:
Insertion sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It avoids the need for recursion
- b. It allows for efficient parallel processing and sorting of data
- c. It only works on small datasets
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **8**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It divides the list into sublists
- b. It recursively sorts subproblems
- c. It uses the heap data structure
- d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **9**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are ignored
b. The parts are merged without sorting
c. Each part is sorted independently
d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **10**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. All of the above
b. Searching
c. Selection
d. Duplicates

The correct answer is: All of the above

Question **11**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
b. A sorting algorithm is stable if it preserves the order of duplicate keys
c. A sorting algorithm is stable if it preserves the order of non-duplicate keys
d. A sorting algorithm is stable if it preserves the order of all keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **12**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. arrange()
- c. order()
- d. sorted()

The correct answer is: sorted()

Question **13**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Dividing the list into halves
- c. Bubbling up the smallest element
- d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **14**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Comparing each element with the others
- c. Combining sorted sublists
- d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **15**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 200,470,80,150,40,90,300,400,70,120

b. 40,70,80,90,120,150,200,300,400,470

c. 40,80,90,150,200,300,400,470,70,120

d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120